

The Morphosyntax of Ezafe in Southern Zazaki*

Songül Gündoğdu
Muş Alparslan University

Arsalan Kahnemuyipour
University of Toronto Mississauga

Sable Peters
University of Toronto

Abstract The present study describes and analyzes the morphosyntactic expression of the Southern Zazaki Ezafe – a linking element in the nominal domain common among Iranian languages. This morpheme is used to link modifiers (i.e. adjectives and possessors) to their head nouns as follows: N-EZ1 MOD1-EZ2 MOD2-EZ3 MOD3. Southern Zazaki, like other languages of the Kurdish region (and unlike, e.g. Persian) reflects phi-features (and case) of the head noun on each Ezafe morpheme in a noun phrase. This paper is focussed around two morphosyntactic puzzles that arise in Southern Zazaki. First, while the Ezafe marker in general reflects the case of the entire DP, the presence of a possessor produces invariant oblique case, regardless of the case value assigned to the DP externally (Paul 2009; Todd 2002; Toosarvandani and van Urk 2014 *i.a.*). Second, Southern Zazaki uniquely employs a separate series of “D-form” Ezafe morphemes in certain syntactic contexts (Todd 2002; Paul 2009; Werner 2018; Keskin 2010 *i.a.*). This study aims to provide a cohesive analysis of Ezafe in Southern Zazaki both with respect to its general phi- and case-sensitive realizations, as well as the distribution of D-forms.

Keywords: Zazaki, Ezafe, Iranian Languages, Case, Nominal Linker

1 Introduction

The Ezafe is a common feature of many Iranian languages wherein a vowel is inserted between the head noun and post-nominal modifiers (including possessors) iteratively (Samiiian 1994; Ghomeshi 1997; Samvelian 2007; Larson and Yamakido 2008; Kahnemuyipour 2014), as indicated in the following schema for a noun phrase with three modifiers:

(1) [N-EZ1 MOD1-EZ2 MOD2-EZ3 MOD3]

The form of the Ezafe morpheme is invariant (barring minor phonological modifications) in some Iranian languages such as Persian, where it appears as *-e* (or *-ye* after a vowel) (2).

* We gratefully acknowledge funding from the Social Sciences and Humanities Research Council of Canada (SSHRC grant #435-2018-0527). Much of the work on this paper was completed when the first author held the Elahé Omidyar Mir-Djalali Postdoctoral Fellowship in Iranian Linguistics at the University of Toronto, generously funded by Roshan Cultural Heritage Institute and the SSHRC grant. We would like to thank the members of the “The Syntax of Nominal Linkers” project and the audience of CLA 2019, 2020 and ICIL 2021 for their valuable feedback on earlier versions of this work. We are truly grateful to Sultan Gümüşkaşık and another anonymous linguistic consultant for generously sharing their knowledge of Zazaki. We also sincerely thank the two anonymous reviewers for their constructive feedback and suggestions. All errors are ours.

- (2) a. *māshin-e sefid-e Ali*
 car-EZ white-EZ Ali
 ‘Ali’s white car’
- b. *xune-ye qashang-e hamsāye*
 house-EZ beautiful-EZ neighbour
 ‘the neighbour’s beautiful house’

In other Iranian languages, such as various dialects of Kurdish, the Ezafe morpheme cross-references the phi-features of the head noun and (sometimes) the case of the noun phrase it appears in. This paper describes and analyzes the morphosyntax of Ezafe in one such language, namely Zazaki.

Zazaki, also known as Dimlî or Kirmanjkî, is a Northwestern Iranian language spoken in Southeastern Turkey (Paul 2009; Werner 2018). It has three main dialects, Southern Zazaki, Central Zazaki and Northern Zazaki (Keskin 2010). This paper focuses on the distribution of Ezafe in Southern Zazaki, although it may occasionally refer to other dialects, when warranted.¹ All Zazaki dialects have a two-case system: Direct (DIR) and Oblique (OBL).² Zazaki exhibits a split ergative system. In the present tense, subjects are marked with the DIR case while direct objects appear in the OBL case. The alignment reverses in past tense sentences where subjects are marked with the OBL and direct objects appear in the DIR case (Paul 2009; Todd 2002). Objects of prepositions and all other nominals excluding locatives and vocatives always appear in the OBL case. Zazaki also displays a two-way gender distinction – feminine and masculine – which is marked on adjectives, verbs and case endings (Todd 2002). As noted above, Zazaki employs the Ezafe marker within the noun phrase and is a language in which the Ezafe morpheme reflects phi-features and case, a phenomenon that is not universal among Iranian languages that use the Ezafe.

The distribution of Ezafe in Southern Zazaki presents several intriguing problems for the morphosyntax of nominals. First, the exact mechanism of case valuation on this linking element is in need of a cohesive solution, in light of an unexpected asymmetry. While the Ezafe marker in general reflects the case features of the noun phrase it appears in, be it DIR or OBL, possessive constructions produce invariant OBL case, even if the noun phrase is in a DIR syntactic environment (e.g. the subject of a present tense sentence or the object of a past tense sentence). Previous approaches (e.g. Toosarvandani and van Urk 2014) have been capable of handling only certain Zazaki varieties, with relatively complex agreement mechanisms. This distribution discussed in the first part of the present paper is found (with minor differences) in all dialects of Zazaki.

There is a further phenomenon of interest unique to Southern Zazaki: in this dialect, in contexts where we have two modifiers (N-EZ1 MOD1-EZ2 MOD2), EZ2 sometimes appears in its expected form as *-a/-o/-ê*, and other times as *-dê/-da* (Todd 2002; Paul 2009; Paul 1998; Werner 2018; Keskin 2010). Previous descriptions of this pattern have referred to these D-forms (*-dê/-da*) simply as “double” Ezafe, or have otherwise analyzed them as case stacking / suffixaufnahme (Larson and Yamakido

¹ All Southern Zazaki data come from the authors’ fieldwork with two native Southern Zazaki speakers from Siverek (Şanlıurfa), currently living in Diyarbakır. We are very grateful to Sultan Gümüşkaşık and another anonymous linguistic consultant for sharing their linguistic knowledge.

² Direct and oblique are established terms used in Iranian linguistics to refer to the cases in a two-cased system where Direct represents Nom/Abs and Oblique all other cases.

2006). In this paper, we review some shortcomings of these accounts and instead present an analysis which takes the D-forms to be instances of grammatically conditioned allomorphy.³

This paper is organized as follows. In sections 2–4, we discuss the case asymmetry seen with OBL Ezafe in possessives, lay out previous solutions, and provide an analysis of Ezafe including its phi and case agreement patterns based on previous well-established syntactic analyses of Ezafe in other Iranian languages (Persian: Kahnemuyipour 2014; Kurdish, Persian: Atlamaz 2016). In section 5, we discuss the D-forms of the Ezafe unique to Southern Zazaki. Section 6 concludes the paper.

2 Morphosyntactic realization of Ezafe in Southern Zazaki

In this section, we describe the morphosyntactic realization of Ezafe in Southern Zazaki. As discussed in the previous section, Ezafe in Southern Zazaki inflects for number, gender, and case. Table 1 shows this paradigm.

Table 1: The forms of the Ezafe morpheme in Southern Zazaki..

	DIR.	OBL.
MASC.	-(y)o	-ê/y
FEM.	-(y)a	-(y)a
PL.	-ê/y	-ê/y

In (3), we can see the sensitivity of the Ezafe vowel to the phi-features of the head noun. We see the feminine form of the Ezafe with the feminine ‘goat’ (3a) and the masculine form with the masculine ‘dog’ (3b). In the plural, the gender distinction is neutralized (3c).

- (3) a. *bız-a sipê* b. *kutık-o sipê* c. *sol-ê sipê*
 goat-EZ.F white dog-EZ.M white shoe-EZ.PL white
 ‘the white goat’ ‘the white dog’ ‘the white shoes’

As noted above, Southern Zazaki Ezafe also inflects for case. In fact, the forms in (3) represent the Direct forms of the Ezafe morpheme. We can see different forms by placing the modified noun in various case positions. When a modified noun is the subject of a transitive clause in the present tense or the direct object in the past tense, i.e. in a position where it receives DIR case, the direct form of the Ezafe is used. The examples in (4)–(6) show this pattern for the subject of a transitive clause in the present tense, where the Ezafe is realized as *-a* with the feminine singular ‘goat’ (4) and as *-o* with the masculine singular ‘dog’ (5). The gender marking is neutralized in the plural (6).

³ In this paper, we are focusing on Ezafe patterns in Southern Zazaki which set it apart from other Iranian languages and are therefore of particular interest to those working on Ezafe and its proper syntactic analysis. For a broader empirical coverage of Ezafe in Zazaki, see Gündoğdu and Bulan (2023).

- (4) *[Biz-a sipê] mun vin-en-a*
goat-EZ.F.DIR white 1SG.OBL see-PRS-3SG.F
'The white goat sees me.'
- (5) *[Kutik-o sipê] mun vin-en-o*
dog-EZ.M.DIR white 1SG.OBL see-PRS-3SG.O
'The white dog sees me.'
- (6) *[Kutik-ê sipê] / [Biz-ê sipê] mun vin-en-ê*
dog-EZ.PL.DIR white / goat-EZ.PL.DIR white 1SG.OBL see-PRS-PL
'The white goats / dogs see me.'

Meanwhile, when the modified noun appears in a context where it receives OBL case, e.g. the direct object of a present verb, the OBL form of the Ezafe is used, as seen in (7)-(9). We can see the contrast between the DIR and OBL forms of the Ezafe in the masculine singular, which are *-o* and *-ê*, respectively (compare (5) and (8)).

- (7) *Ez-a [biz-a sipê] vin-en-a*
1SG.DIR-PROG.1SG goat-EZ.F.OBL white see-PRS-1SG
'I see the white goat.'
- (8) *Ez-a [kutik-ê sipê] vin-en-a*
1SG.DIR-PROG.1SG dog-EZ.M.OBL white see-PRS-1SG
'I see the white dog.'
- (9) *Ez-a [kutik-ê sipê] / [biz-ê sipê] vin-en-a*
1SG.DIR-PROG.1SG dog-EZ.PL.OBL white / goat-EZ.PL.OBL white see-PRS-1SG
'I see the white dogs / the white goats.'

The oblique form of the Ezafe is also used in the context of a possessive construction (10). The Ezafe forms used in this context are the same as those used in the clausal oblique positions in (7)-(9). Once again, the only distinctive form of the Ezafe is the feminine (10a) as the masculine and plural forms (10b-10c) are syncretic (see Table 1).

- (10) a. *biz-a Ahmad-∅ / Sultan-e / mun*
goat-EZ.F.OBL Ahmad-OBL.M / Sultan-OBL.F / 1SG.OBL
'Ahmad's / Sultan's / my goat'
- b. *kutik-ê Ahmad-∅ / Sultan-e / mun*
dog-EZ.M.OBL Ahmad-OBL.M / Sultan-OBL.F / 1SG.OBL
'Ahmad's / Sultan's / my dog'
- c. *sol-ê Ahmad-∅ / Sultan-e / mun*
shoe-EZ.PL.OBL Ahmad-OBL.M / Sultan-OBL.F / 1SG.OBL
'Ahmad's / Sultan's / my shoes'

Importantly, when a possessive construction is used in the clause, the form of the Ezafe remains oblique, regardless of whether it is in an oblique position (11a) or a direct position (11b). This is a key issue we aim to account for in this paper.

- (11) a. *Ez-a* [*kutik-ê* *to*] / [*biz-a* *to*]
 1SG.DIR-PROG.1SG dog-EZ.M.OBL 2SG.OBL / goat-EZ.F.OBL 2SG.OBL
 vin-en-a
 see-PRS-1SG
 ‘I see your dog / your goat.’
- b. [*kutik-ê* *to*] / [*biz-a* *to*] *min* *vin-en-o/a*
 dog-EZ.M.OBL 2SG.OBL / goat-EZ.F.OBL 2SG.OBL 1SG.OBL see-PRS-3SG.M/F
 ‘Your dog / goat sees me.’

In this section, we have looked at the morphosyntactic realization of the Ezafe in Southern Zazaki. We have seen that the Ezafe morpheme invariably tracks the phi-features of the head noun. In addition, the Ezafe is sensitive to the case features of the DP it is part of, showing a contrast between the direct case environment (the subject of a transitive clause in the present and the direct object of a transitive clause in the past) and the oblique case environment (the object of a transitive clause in the present and the subject of a transitive clause in the past). The oblique form of the Ezafe is also used in the context of a possessive construction. Meanwhile, in this context, the Ezafe morpheme still tracks the phi-features of the head noun (as opposed to the possessor). When the possessive construction is placed in the clausal context, the oblique form of the Ezafe persists even when used in the direct case environment. Before providing our analysis of the pattern presented in this section, in the following section, we review Toosarvandani and van Urk’s (2014) account of the same pattern, which uses bi-directional Agree. This is the only formal account of these facts in the literature, as far as we are aware.

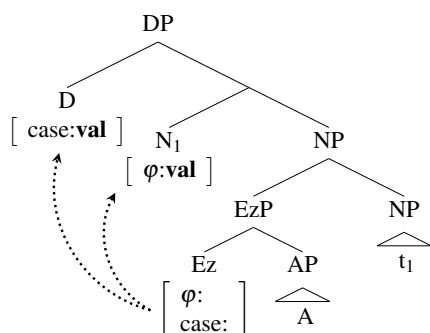
3 A Bi-directional Agree Account (Toosarvandani and van Urk 2014)

Toosarvandani and van Urk (2014) (hereafter TvU) analyze the distributional pattern described in section 2 as a result of bi-directional Agree. In their system, there is a preference for downward Agree, but when no such relation can be established, upward Agree (Pesetsky and Torrego 2007; Zeijlstra 2012; Bjorkman and Zeijlstra 2019 etc.) becomes available. The syntax TvU assume for the Ezafe construction is one in which the Ezafe consistently takes the noun modifier (be it an adjective or a possessor) as a complement. Under this view, this EzafeP is adjoined to NP, as we will see in more detail below.

Let us first consider agreement in the context of adjectives (with no possessor). TvU’s structure for this option is given in (12). In this context, the Ezafe, which has unvalued phi and case features, takes AP as its complement. No Agree relation can be established between Ez and AP because AP does not have any inherent case or phi features. Then, EzP adjoins to the NP, which contains the N head with inherent phi features. Meanwhile, no Agree relation can be established between Ez and N, as N is not in its search space. Next, N is copied and re-merged in a position above

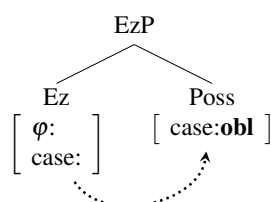
the Ezafe⁴, followed by the introduction of D into the structure. D gets its case value externally depending on the position the DP finds itself in, be it DIR or OBL. Crucially, phi- and case on Ez are valued via an upward Agree relation with N and D, respectively, as shown in (12).

(12) Toosarvandani and van Urk 2014, p. 6



According to TvU, in the context of a possessive construction, Ezafe merges with the possessor carrying an oblique case, as shown in (13). Consequently, given the preference for downward Agree, Ezafe establishes an Agree relation with the possessor and copies its Oblique case feature.

(13) Toosarvandani and van Urk 2014, p. 7

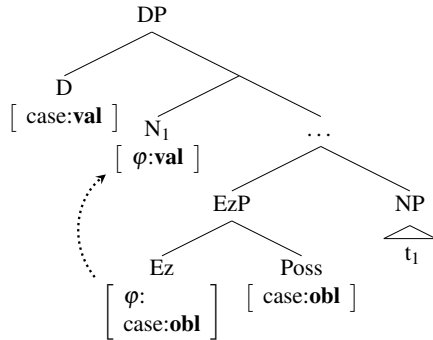


This raises a question as to why the phi-features of Ez are not valued as a result of the same Agree relation. We know from section 2 that the phi-features on Ez are still valued by the head noun and not the possessor. TvU posit that this is due to the fact that there is a null P which assigns Oblique case to the possessor and that this P is phasal, making the phi features of the possessor inaccessible for Agreement as a result of the Phase Impenetrability Condition (PIC)⁵. As a result, Ezafe probes upward and values its phi with the phi features of N, in a manner similar to the previous adjectival context. This is shown in (14). Crucially, the Case feature on Ez is already valued and unlike the adjectival context shown in (12), it does not get valued by the case feature on D.

⁴ Toosarvandani and van Urk (2014) do not discuss the details of this movement or the syntactic position N moves to.

⁵ It is not clear why the same PIC mechanism does not prevent the valuation of case features on EZ by the possessor.

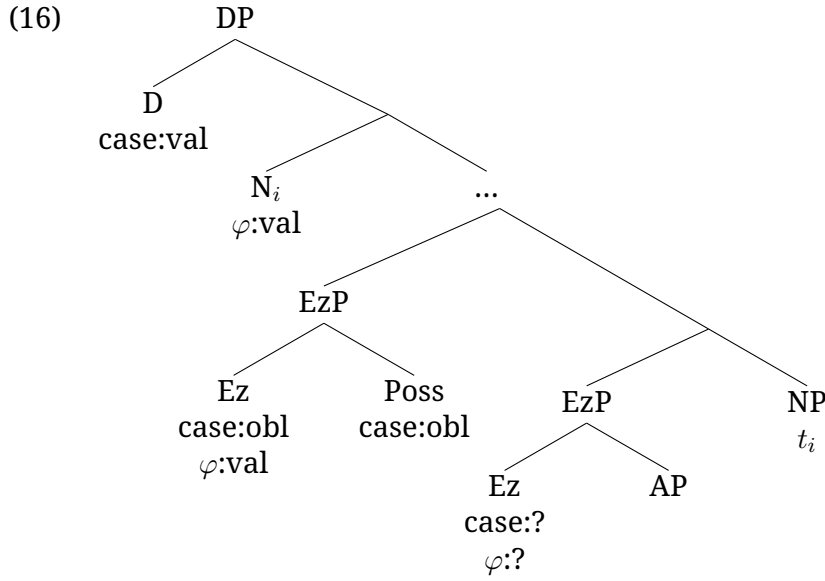
(14) Toosarvandani and van Urk 2014, p. 8



This analysis leaves open several questions with respect to the analysis of Ezafe in general, as well as to the specific mechanisms underlying the Zazaki case. For instance, it is unclear under their analysis how structures with multiple modifiers (e.g. multiple adjectives, or an adjective and a possessor) are handled. Recall that in TvU's system, each Ezafe takes its corresponding noun modifier as a complement. Thus, it is reasonable to assume that with multiple modifiers, each of the EzPs would be adjoined to the NP in the right order, with the N moving to a position under D. A point needs to be made with respect to the correct order of modifiers, when one of them is an adjective and the other a possessor. In Southern Zazaki (as well as other dialects of Zazaki), the possessor goes before the adjective: N-Ez Poss'r-Ez A. This order, which is also found in Kurmanji (Northern Kurdish), is shown in the Southern Zazaki example in (15a). The more common order in Iranian languages is one which has the adjective before the possessor. This order is shown for Persian in (15b).

- (15) a. *şit-ê manga-yê sipê*
 milk-EZ.M.OBL cow-EZ.M.OBL white
 'the cow's white milk' Southern Zazaki
- b. *shir-e sefid-e gāv*
 milk-EZ white-EZ cow
 'the cow's white milk' Persian

Extending the logic of TvU's analysis to the example given in (15a) leads to the structure in (16).



Meanwhile, the structure in (16) would predict that only the Ezafe introducing the possessor would get its case valued as OBL by its complement, namely the possessor. The Ezafe introducing the adjective should not be able to get its case valued by the possessor and should instead have its case feature valued by upward Agree with D, which may be DIR, if placed in the right syntactic position. This prediction is not borne out, as seen in (15a) where both Ezafes are marked as OBL consistently (regardless of whether the whole DP is in a DIR or OBL position).⁶

There are other questions with respect to the above analysis, for example the merge position of the head noun and more importantly its movement to a position above the modifiers. What kind of position is this? Is it head movement? Is it movement to the specifier of a different functional projection? What is this projection? What motivates this movement? These questions are not addressed by TvU. In the next section, we develop an account of Ezafe in Southern Zazaki based on the phrasal movement analysis of Kahnemuyipour (2014), developed for Persian, and Atlamaz's (2016) modification to capture the aforementioned order difference between Persian and Kurmanji (which has the same order as Southern Zazaki).

4 The Morphosyntax of Southern Zazaki Ezafe: A new analysis

We have shown so far that the Southern Zazaki Ezafe marker shows sensitivity to the phi-features (both number and gender) of the head noun as well as the case features of the noun phrase it appears in, be it DIR or OBL. Meanwhile, when the Ezafe appears in a possessive construction, it appears in an OBL form, even if the noun phrase is in a DIR syntactic environment, namely the subject of a present tense sentence or the object of a past tense sentence. In this section, we provide an analysis of the morphosyntactic realization of Ezafe in Southern Zazaki.

⁶ In fact, in Northern and Central Zazaki, in an example such as (15a), it is only the Ezafe preceding the possessor that consistently appears in the OBL form. The case value of the second Ezafe depends on the syntactic context of the whole DP, and takes the DIR form if used in the DIR context. TvU's analysis does not encounter any problems in accounting for the case pattern in Central or Northern Zazaki. In section 4, we propose a new analysis which can account for the Southern Zazaki pattern in (15a). In footnote 16, we discuss how a minor variation to the Southern Zazaki system can capture the Central and Northern Zazaki facts.

This type of feature matching found within the noun phrase, exemplified by the case of Southern Zazaki Ezafe, is typically referred to as Nominal Concord (Cinque 1994; Carstens 2000; Baker 2008a; Baier 2015; Norris 2017a; Norris 2017b *inter alia*). The features most commonly involved in Nominal Concord are phi-features and case, as is the case with Southern Zazaki Ezafe. Different analyses have been provided for phi-feature Concord, but they can be largely divided into two camps. One camp takes nominal concord to be an instantiation of the operation Agree, also responsible for subject-verb agreement (see, for example, Bonet 2013; Carstens 2001; Carstens 2011; Carstens 2016; Cinque 1994; Collins 2004; Danon 2011; Baier 2015; Baker 2008b; Kramer 2009; Landau 2016; Toosarvandani and van Urk 2014). Another approach takes nominal concord to be crucially different from subject-verb agreement and as such obtained via a different mechanism (see, for example, Polinsky 2016; Norris 2014; Norris 2017b; Giusti 2008). In particular, Norris (2014; 2017) outlines several systematic differences between nominal concord and subject-verb agreement and develops an alternative analysis for nominal concord which consists of two components, a syntactic feature percolation mechanism and a morphological feature copying on Agr nodes inserted post-syntactically. In this paper, we provide an Agree-based analysis of phi-feature concord in Zazaki. This choice is mainly for ease of illustration due to the familiarity of the Agree mechanism. We do not anticipate any serious problems for a Norris-style feature percolation and feature copying analysis of these facts. It should be noted that the most important problem with some of the other Agree analyses of nominal concord, namely the absence of a c-command relation between the concord element and the head noun does not extend to the case of Southern Zazaki Ezafe, as we will see below. We will argue that Southern Zazaki Ezafe carries unvalued phi-features, which are valued via an Agree relation with the interpretable phi-features on the head noun, in a manner discussed in detail below⁷.

There is less controversy over the analysis of Case Concord in the literature. Most analyses of case take it to be assigned to a maximal projection, hence arising in D (rather than N). In case concord, this case feature is taken to spread downward to the other elements inside the DP (see, for example, Babby 1987; Chomsky 1981; Chomsky 1986; Delsing 1993; Halpert 2015; Norris 2018)⁸.

There are very few analyses which provide an Agree-based account of case concord (see, for example, Harley 2013 and Toosarvandani and van Urk 2014). These authors need to resort to Upward Agree (Pesetsky and Torrego 2007; Zeijlstra 2012; Bjorkman and Zeijlstra 2019 etc.) in this context, as the unvalued case features on concord elements inside the noun phrase need to probe upward to get the case value on D. Norris (2014; 2017) extends his analysis of phi-feature concord to case concord, with the only difference that the feature percolation is downward with case concord and not upward, as is the case for phi-feature concord. In this paper, we do not take a strong position on whether case concord is the result of Upward Agree or a Norris-style downward feature percolation and feature copying. Either

⁷ Southern Zazaki Ezafe carries an unvalued case feature as well, which is also involved in concord. As we will see below, it is really the Agr heads enabling the phrasal movements involved in the derivation of the Ezafe construction which carry the phi- and case features. In other words, what we call the Ezafe is really the realization of these Agr heads with phi- and case features valued in a manner detailed below.

⁸ There are some approaches which do not take the DP as the intermediary for case assignment and take case to be assigned directly to the various elements (Brattico 2008; Brattico 2010; Brattico 2011; Matushansky 2008; Richards 2013).

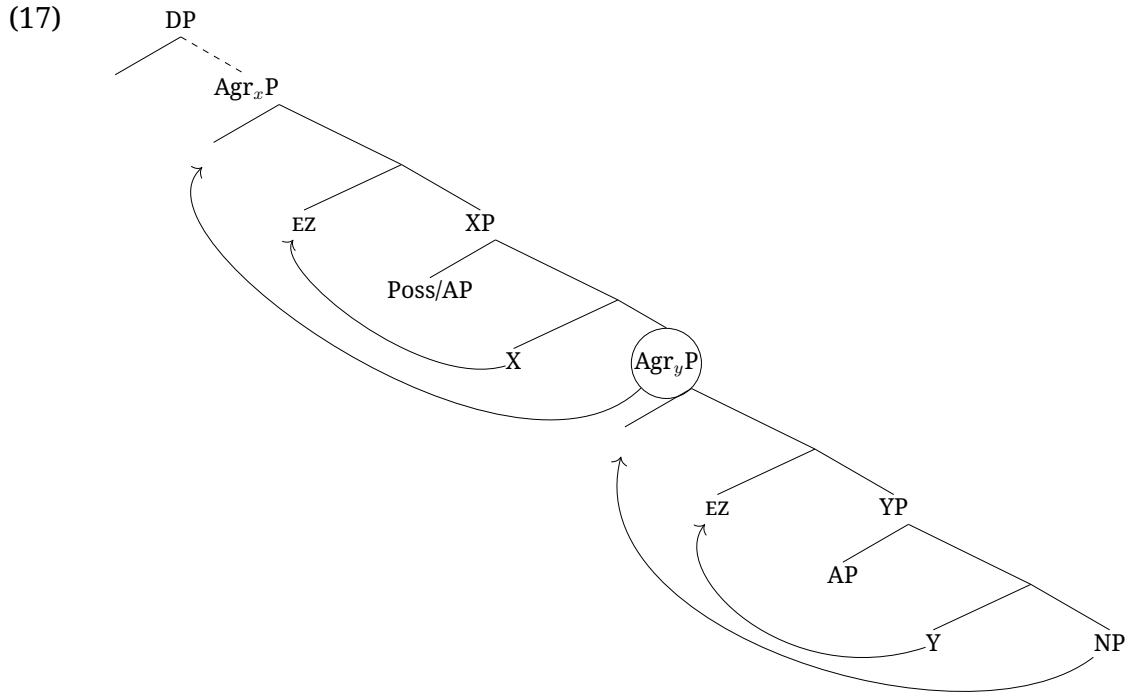
of these mechanisms should be able to account for the Southern Zazaki case concord facts. In our diagrams, we will simply show it as case concord. This should not be taken to imply a fundamental difference between phi-feature and case concord.

Let us review the realization of case in Southern Zazaki briefly before looking at the structural account in more detail. When there is no possessive involved, D gets its case value externally based on the structural position in which the DP appears, namely DIR (subject of a transitive clause in the present, or object of a transitive clause in the past) or OBL (object of a transitive clause in the present, or subject of a transitive clause in the past). In the context of a possessive construction, the case feature on D is valued internally as genitive, which is morphologically realized as OBL in the language. Once valued internally, the case feature on D cannot be revalued externally, and consequently the OBL feature persists on D even when the DP is placed in a DIR position. Crucially, the case feature on D is spread onto all the Ezafe markers inside the DP via Concord. In what follows, we work out the details of this proposal.

With respect to the syntax of the Ezafe construction, we follow Kahnemuyipour's (2014) phrasal movement analysis of this construction in Persian, informed by Atlamaz's (2016) conclusions with respect to height and order of adjunction within the DP.⁹ Let us start with a brief overview of Kahnemuyipour's (2014) proposal. According to this analysis, the noun phrase in Ezafe languages is taken to be head-final, with the modifiers residing in the specifiers of projections above N. This structure is in line with other roll-up analyses of DP structure in other languages within the framework best known as cartography (Cinque 2002; Cinque 2005; Cinque 2010; Shlonsky 2004; Shlonsky 2010 *inter alia*, but see Abels and Neeleman (2012) for a different perspective.). In addition to the projections which house the modifiers, there are intermediate projections which enable the roll-up derivation. The relevant structure and roll-up movements are shown schematically in (17), where the projections hosting the modifiers are marked as XP, YP, etc. and the intermediate projections are marked as AgrPs (in line with the literature on roll-up movement). Under this view, the Ezafe marker can be seen as the surface realization of the suggested inversion process on Agr heads, i.e. akin to a linker in the sense of den Dikken (2006).¹⁰

⁹ There is an ongoing debate with respect to the proper analysis of Ezafe and similar elements in Iranian languages. While we have adopted Kahnemuyipour's (2014) roll-up analysis, other scholars have analyzed Ezafe as a case assigner (Samiiian 1983; Samiiian 1994; Larson and Yamakido 2008; Larson and Samiiian 2020), a PF element inserted to maintain the phrasehood of a nominal (Ghome-shi 1997; Ghaniabadi 2010), a phrasal affix that marks the presence of a modifier or complement (Samvelian 2007; Samvelian 2008), an element that converts AdjPs from property-denoting expressions to propositions (Y. Karimi 2007). An alternative style of analysis (Kahnemuyipour and A. Peters 2019) views the Ezafe as an element inserted to resolve a labeling problem (Chomsky 2013) that arises from symmetrical constructions for modification / adjunction that arise in certain languages (i.e. those of the Iranian type). We do not aim to address all these analyses in the context of this paper and refer to specific works only when the Zazaki data have direct implications for those analyses.

¹⁰ Support for the idea that Ezafe is the realization of Agr-heads comes from Zazaki and other Iranian languages with Ezafe (with the exception of Persian) in which phi-feature agreement is reflected on the Ezafe marker.

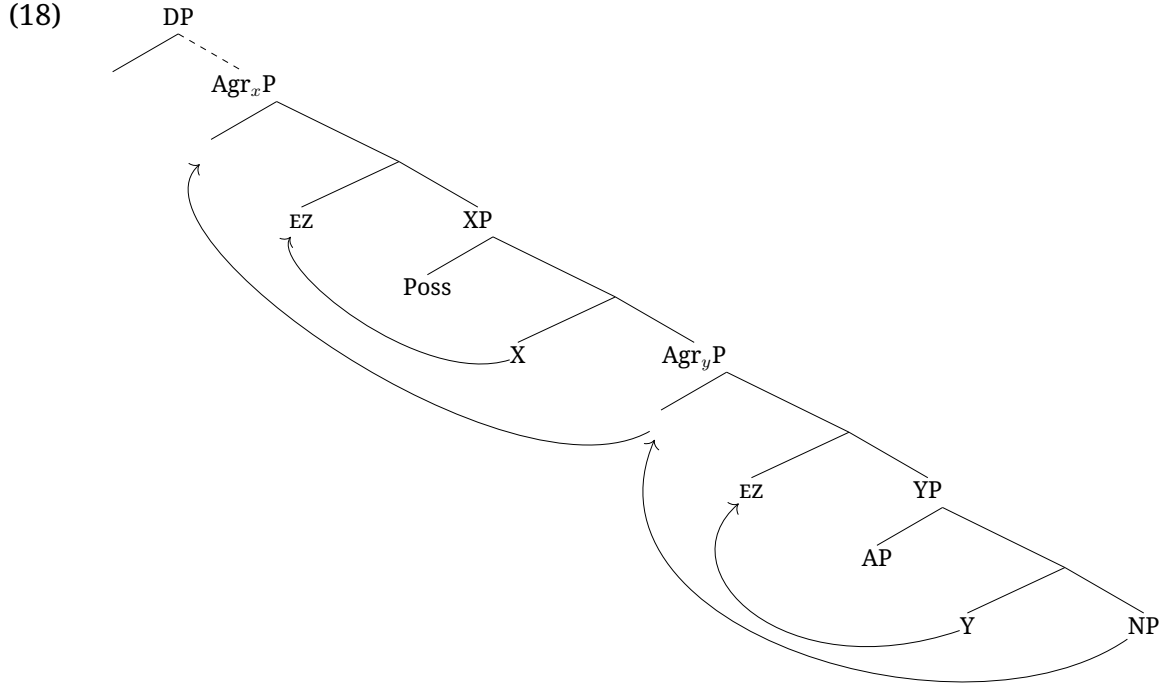


With the analysis shown in (17), if as standardly assumed, possessors are merged higher than adjectives, the expected linearization of modifiers is one in which the adjective precedes the possessor. This is because roll-up derivations lead to a mirror-image order of the base-generated constituents. This is precisely the order we find in Persian (15b) (and several other Iranian languages such as Sorani Kurdish), which Kahnemuyipour's analysis was proposed for. As discussed in Atlamaz (2016) for Kurmanji, and noted above for Southern Zazaki, the order in these languages is different, with the possessor appearing before the adjective (15a).¹¹

To capture this order difference, Atlamaz (2016) suggests that while in languages like Persian, we are dealing with roll-up movement, in languages like Kurmanji (and Southern Zazaki), no roll-up movement ensues and instead the NP moves alone through the specifiers of the intermediate projections. This analysis, which is what we are adopting for Southern Zazaki, is shown schematically in (18).¹²

¹¹ It should be noted that the Persian order (N-EZ A-EZ Poss) is possible in Southern Zazaki but it has a contrastive reading. This alternative order may be the result of focus-movement. We are abstracting away from this possible order and focusing on the neutral order of elements.

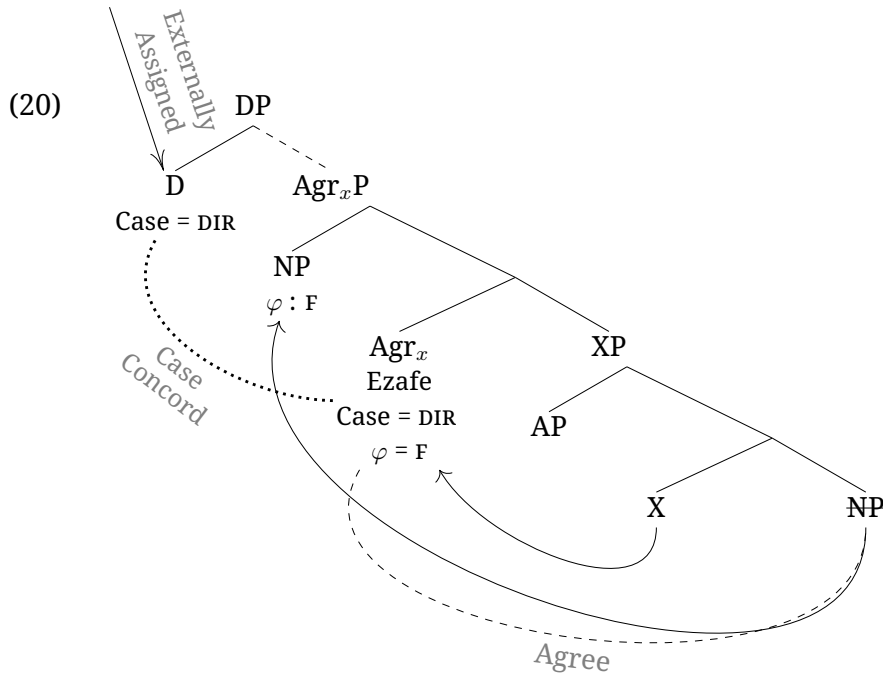
¹² As noted by Atlamaz (2016), this proposal is very similar to Cinque's (2005) approach to capture word order differences inside noun phrases across languages. According to Cinque, the roll-up movement option, where the NP pied-pipes other material with it is less marked than the movement of the NP alone, but the latter is still observed in some languages.



Building on the analysis schematized in (18), we first account for the Ezafe pattern in the context of adjectives. The structure for the simple noun phrase used in the example repeated in (19) (inside square brackets) is given in (20). We assume that NPs are merged with inherent phi-features. The AP is merged in the specifier position of XP just merged above NP. Then the Agr_x head is merged which contains unvalued phi- and case features, as shown in (20). An Agree relation is established between Agr_x and the NP, valuing its phi-features as feminine and singular. The head X moves to Agr_x , followed by the movement of the NP to Spec Agr_x P, resulting in the reverse order. Then D is merged with an unvalued Case feature which gets valued externally as DIR. Finally, the case feature on the Agr_x , gets valued as DIR as a result of case concord with D.¹³ The Ezafe is then realized morphologically as *-a*, the singular feminine DIR form.

- (19) [*Bız-a sipê*] *min vin-en-a*
 goat-EZ.F.DIR white 1SG.OBL see-PRS-3SG.F
 ‘The white goat sees me.’

¹³ As noted above, Case concord can be the result of upward agree or a Norris-style downward feature percolation and feature copying. Nothing in our proposal hinges on this analytical choice.

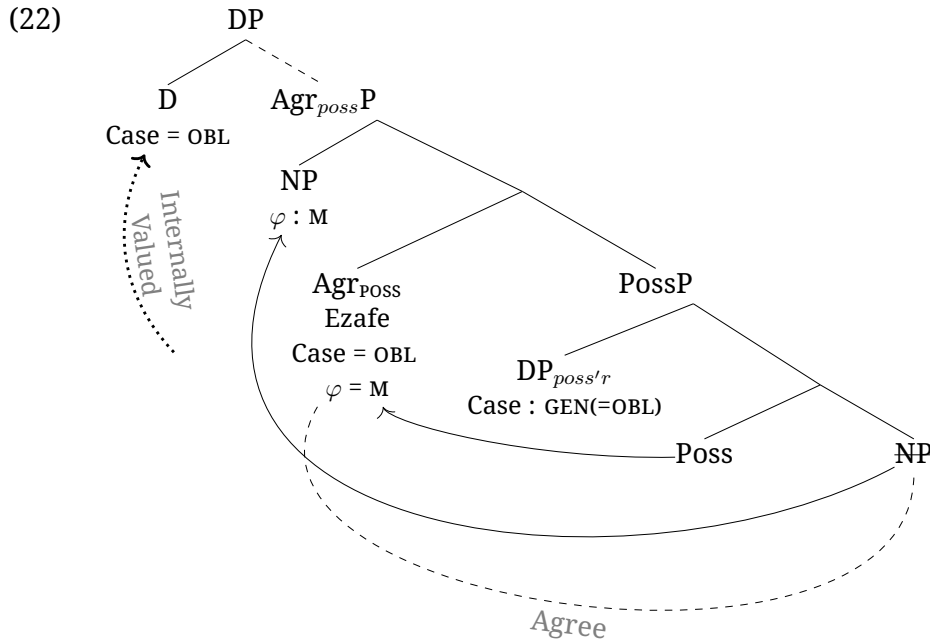


Turning to the Ezafe used in the possessive context, let us consider the example (21), with the possessive construction used as the subject of a transitive clause used in the present – a DIR case context. Recall that in the possessive context, the case on the Ezafe is OBL regardless of whether the DP is placed in an OBL or DIR context. The structure for the possessive construction (inside square brackets) in (21) is given in (22). Once again, we assume that the NP is merged with inherent phi-features. For the possessor, instead of using labels such as XP and YP, we follow standard convention and take the functional projection to be PossP. Crucially, the Poss head carries a GEN case feature, which is realized as OBL in Zazaki. The possessor which is merged in SpecPossP, has its case valued as GEN(=OBL) via the Poss head. The GEN(=OBL) marking on the possessor was illustrated in the examples in (10). In (21), we can see the pronominal form *to*, which is used in the possessive and other OBL contexts. The possessor is merged in the specifier of PossP. Then Agr_{Poss} head is merged which contains unvalued phi- and case features. An Agree relation is established between Agr_{Poss} and the NP, valuing its phi-features as masculine and singular.¹⁴ There is also an Agree relation between Agr_{Poss} and Poss, which values its case-feature as GEN (=OBL). This leads to the realization of the Ezafe as the singular masculine oblique form *ê* in the morphology. The Poss head moves

¹⁴ An anonymous reviewer raises a question as to why the possessor does not intervene in this Agree relation, leading to phi-features on Agr_{Poss} being valued by the possessor. There are different ways of accounting for lack of intervention in this context. Here, we tentatively attribute this to the activity condition (Chomsky 2000; Chomsky 2001 and subsequent authors). Having had its case valued, the possessor is deemed inactive and unable to establish an Agree relation with the Agr head. Another way to handle this absence of intervention is to assume the presence of an additional probing feature such as N, which is present on NP but not the possessor (see Cinque 2005 for such a solution). It is worth noting that Norris 2014 uses the lack of intervention by the possessor in phi-feature concord as an argument against Agree-based analyses of this phenomenon. Crucially, his proposed (upward) phi-feature percolation process does not interact with the possessor at all.

to Agr_{Poss} followed by the movement of NP to $\text{SpecAgr}_{\text{Poss}}\text{P}$. Then the D is initially merged with an unvalued Case feature, which is subsequently valued internally as OBL. When this DP is merged in the subject position of the clause, the OBL case on the DP persists despite the DIR context of the whole DP. This is to say that once the case on D is valued internally, external case marking cannot overwrite its case value.¹⁵

- (21) *[Kutik-ê to] min vin-en-o*
 dog-EZ.M.OBL 2SG.OBL 1SG.OBL see-PRS-3SG.M
 ‘Your dog sees me.’

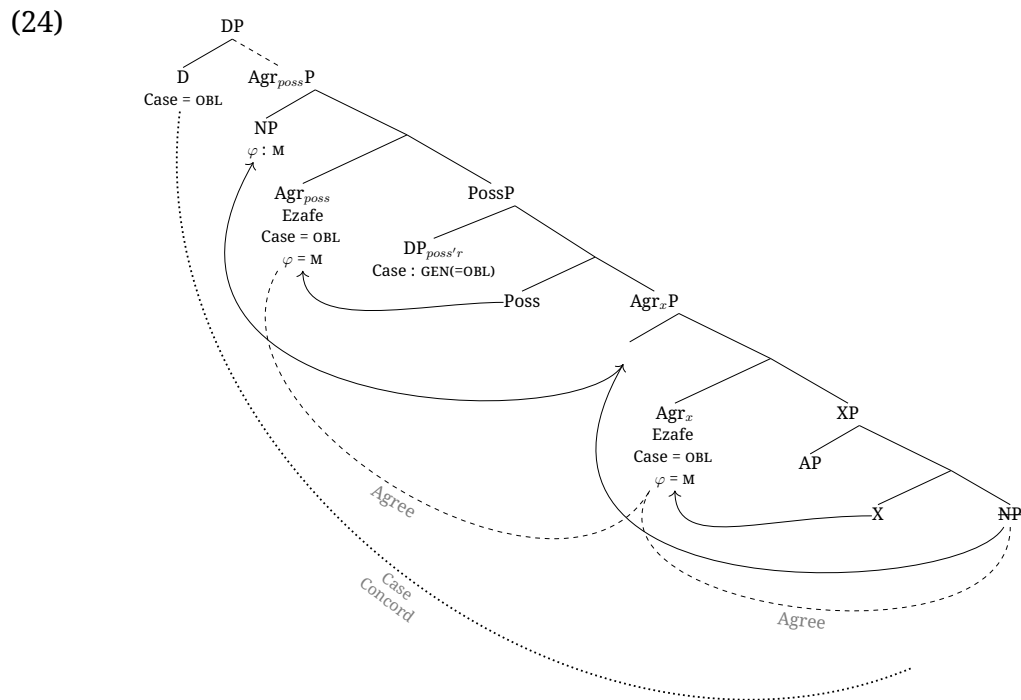


Having considered separately the contexts of a noun modified by an adjective and one used in a possessive construction, we can now turn to an example which combines the two. One such example is repeated from (15a) in (23). The syntactic derivation for this example is given in (24). Once again, the NP is merged with inherent phi-features, in this case masculine singular. The AP is merged in the specifier of XP just above NP. Then Agr_x is merged with unvalued phi and case features. Its phi-features are valued as masculine singular as a result of an Agree relation with the NP. The X head moves to Agr_x and the NP moves over the AP and lands in the specifier of Agr_xP . Then PossP is formed above Agr_xP , hosting the possessor in its specifier. The Poss head carries a $\text{GEN}(=\text{OBL})$ feature, which values the case

¹⁵ An anonymous reviewer raises a question as to why the case on D cannot be overwritten after it has been valued internally. We note that this pattern is found in some other languages (e.g. English (Bejar and Massam 1999), Persian (S. Karimi 2008) etc.), where the case valuation of noun phrases blocks further valuation by other case assigners. We refer the reader to the analysis of Multiple Case Checking in Bejar and Massam (1999), where this phenomenon is only made possible in languages where case is assigned compositionally to a nominal at one end of a chain, and does not move along with said noun. MCC is then impossible in those languages that carry case features along with N during movement. Under this view, Zazaki can be seen as an instance of the latter type. We leave this avenue open for further research.

feature on the possessor as GEN(=OBL). Then Agr_{Poss} is merged carrying unvalued phi-and case features. Its phi-features are valued as masculine singular, via an Agree relation with Agr_x (which has itself established an Agree relation with NP). The case feature is valued as GEN(=OBL) via an Agree relation with Poss. The Poss head moves to Agr_{Poss} and NP moves over the possessor into the SpecAgr_{Poss}P. Then the D head merges with an unvalued case feature, which gets valued internally as OBL via an Agree relation with Agr_{Poss}. Finally, the case feature on D is spread onto Agr_x via Case Concord.

- (23) *şit-ê manga-yê sipê*
 milk-EZ.M.OBL cow-EZ.M.OBL white
 ‘the cow’s white milk’



As a result of the process visualized in (24), both Ezafes in Agr_{Poss} and Agr_x will end up with masculine singular oblique values, as expected.¹⁶

¹⁶ As noted in footnote 6, in Central and Northern Zazaki, in the context of a possessor and an adjective, it is only the Ezafe preceding the possessor which takes the oblique form, and the one preceding the adjective will carry the case value assigned to D externally, DIR if the DP is the subject of a present tense sentence or the direct object of a past tense sentence or OBL if it is the direct object of a present tense sentence or the subject of a past tense sentence. Crucially, the case value on the Ezafe preceding the adjective still reflects the case value on D. This contrasts with Southern Zazaki which exhibits OBL on all the Ezafes. Let us recall how the OBL case on all Ezafes was obtained in Southern Zazaki. There were three steps. The OBL on Agr_{Poss} is the result of an Agree relation with the Poss head. Then, once D is merged, the Case feature on D is valued as OBL via Agree with Agr_{Poss}. This OBL value is extended to the other Ezafes as a result of Case Concord. In order to account for the Central and Northern Zazaki pattern, we simply have to assume that there is no Agree relation between D and Agr_{Poss} in these variants. As a result, D gets its case externally, which can be DIR in the right context. This case value is spread to the other Ezafes as a result of Case Concord, except for the already valued

In this section, we developed an account of the morphosyntax of the Southern Zazaki Ezafe. Building on Kahnemuyipour’s (2014) and Atlamaz’s (2016) analyses of the Ezafe construction, we showed how the Ezafe is the realization of the phi- and case features on intermediate Agr heads which enable the phrasal movement derivation. These phi- and case features receive their values via nominal concord inside the noun phrase. In the next section, we turn to D-forms of Ezafe (sometimes referred to as “double” Ezafe) found uniquely in Southern Zazaki.

5 D-forms of Ezafe in Southern Zazaki

Abstracting away from minor phonological differences, all dialects of Zazaki exhibit the Ezafe patterns discussed in the previous sections. Meanwhile, Southern Zazaki Ezafe has an additional unique property, not found in Central or Northern Zazaki. In Southern Zazaki, in contexts where we have two modifiers, N-EZ1 MOD1-EZ2 MOD2, the EZ2 sometimes appears in its expected form as *-a/-o/-ê*, as discussed above, and other times as *-dê/-da*. The appearance of these D-forms has been noted in the literature (Todd 2002; Paul 1998; Paul 2009; Keskin 2010; Werner 2018) and is sometimes referred to as “double” Ezafe (Larson and Yamakido 2006). In what follows, we lay out the distribution of the D-forms of Ezafe in Southern Zazaki and provide an account for this distribution.

5.1 Distribution of D-forms in Southern Zazaki

One of the contexts in which EZ2 (in the N-EZ1 MOD1-EZ2 MOD2 schema) appears as *-dê/-da* (marked invariantly as DE in the glosses below) in Southern Zazaki is when a modified noun is used as a possessor in the possessive construction: [N_{possessum}-EZ1 [N_{possessor}-EZ2 A]]. Some examples of this kind are given in (25) below. Take (25a). In this example, “white goat” is the possessor and “ear” is the possessum. The Ezafe appearing on the possessum shows the expected form, namely masculine oblique, as “ear” is masculine and its OBL case is due to its use in the possessive construction (see above). What is new here is the form of the second Ezafe which, instead of the usual feminine form *-a*, appears as *-da*. Similarly, in examples (25b) and (25c), we find *-dê* in the same position. The variation between *-dê* and *-da* reflects the phi-features and case of the possessor, where in the GEN(=OBL) context, the latter is used with the feminine singular “goat” (25a) and the former with the masculine singular “dog” (25b) and the plural “friends” (25c). Once again, we can see that the gender distinction is neutralized in the plural.

- (25) a. *goş-ê bız-er-da sıpê*
 ear-EZ.M.OBL goat-OBL.F-DE white
 ‘the white goat’s ear’
- b. *goş-ê kutık-Ø-dê sıpê*
 ear-EZ.M.OBL dog-OBL.M-DE white
 ‘the white dog’s ear’

Case on the Ezafe preceding the possessor, which cannot be overwritten. No other difference needs to be assumed between the two systems. Other aspects of the system presented in this paper for Southern Zazaki apply identically to Central and Northern Zazaki.

- c. *sol-ê* *embaz-an-dê* *min*
 shoe-EZ.PL.OBL friend-OBL.PL-DE 1SG.OBL
 ‘my friends’ shoes’

Another context where EZ2 is realized as *-dê/-da* is the recursive possessive constructions where $[N_{\text{possessum}}\text{-EZ } N_{\text{possessor}}]$ is itself a possessor in another possessive construction, $[N_{\text{possessum}}\text{-EZ1 } [N_{\text{possessum}}\text{-EZ2 } N_{\text{possessor}}]]$. This is shown in the examples in (26). In (26), EZ1 reflects the phi-features of the masculine singular/mass nouns “ear” or “milk” and OBL case due to the possessive context. Crucially, EZ2 appears as a D-form, with *-dê* used with the masculine singular “dog” (26a) or plural “goats” (26c) and *-da* used with the feminine singular “goat” (26b).

- (26) a. *goş-ê* *kutık-Ø-dê* *min*
 ear-EZ.M.OBL dog-OBL.M-DE 1SG.OBL
 ‘my dog’s ear’
 b. *şit-ê* *bız-er-da* *Sultan-e*
 milk-EZ.M.OBL goat-OBL.F-DE Sultan-OBL.F
 ‘Sultan’s goat’s milk’
 c. *şit-ê* *bız-an-dê* *min*
 milk-EZ.M.OBL goat-OBL.PL-DE 1SG.OBL
 ‘my goats’ milk’

At first glance, it may appear that generally in the context of possessive constructions, when one of the nouns is modified, the second Ezafe appears as one of the D-forms. However, as illustrated in the contrast in (27), EZ2 appears as a D-form only if the adjective modifies the possessor (27a) and not when it modifies the head noun (the possessum) (27b) (Paul 1998; Paul 2009; Todd 2002).¹⁷

- (27) a. $[N_{\text{possessum}}\text{-EZ1 } [N_{\text{possessor}}\text{-EZ2 } A]]$
şit-ê *manga-r-da* *siya*
 milk-EZ.M.OBL cow-OBL.F-DE black
 ‘the black cow’s milk’
 b. $[N_{\text{possessum}}\text{-EZ1 } N_{\text{possessor}}\text{-EZ2 } A]$
şit-ê *manga-yê* *sipê*
 milk-EZ.M.OBL cow-EZ.OBL.M white-M.OBL
 ‘the cow’s white milk’

¹⁷ Recall, as mentioned above in example (15a), that in Zazaki (unlike e.g. in Persian), the possessor appears between the head noun and any other modifiers of the head noun, e.g. adjectives (see figure 24 for the derivation). As discussed in footnote 11, the opposite (Persian-like) order is also possible, in contexts where disambiguation is necessary, as in example (i) below. The D-form is not used in this order either.

- (i) $[[N_{\text{possessum}}\text{-EZ1 } A]\text{-EZ2 } N_{\text{possessor}}]$
şit-ê *sipê-yê* *manga*
 milk-EZ.M.OBL white-EZ.M.OBL cow
 ‘the cow’s white milk’

A similar contrast can be found in the context of recursive possessions when we compare a possessive construction used as the possessor to one where it is used as the possessum. The latter type of construction is much less common, but can be found in examples such as [Roald Dahl's [children's books]]. One such contrast is shown in (28). The example in (28a), repeated from (26b), shows a possessive construction used as a possessor: the milk of Sultan's goat; in this example, the D-form is used. In (28b) however, we see a possessive construction placed in the possessum position, where *Sütaş* is a Turkish milk manufacturer, 'goat's milk' being a product of theirs.¹⁸ Here, we can see that the D-form is used only when the Ezafe construction (in this case a possessive) is itself used as the possessor (as opposed to the possessum).

- (28) a. [*şit-ê* [*bız-er-da* *Sultan-e*]]
 milk-EZ.M.OBL goat-OBL.F-DE Sultan-OBL.F
 'Sultan's goat's milk'
- b. [*şit-ê* *bız-an-ê* *Sütaş-e*]]
 milk-EZ.M.OBL goat-OBL.PL-EZ.M.OBL Sütaş-OBL.F
 'Sütaş' goat milk'

All the examples of D-forms discussed so far involve possessive constructions, either a modified noun or a possessed noun itself is used as a possessor. This shows that the existence of a genitive case relation with the head noun is crucial for the appearance of the D-forms. We can, therefore, generalize the –D-forms as the realization of Ezafe (on a modified or possessed noun) whenever it receives genitive case as the possessor. This generalization covers the examples of *-dê/-da* we have seen so far (25)–(28). It also explains why we do not see the appearance of *-dê/-da* in (27b) & (28b) (as opposed to (27a) & (28a)), because in these examples, the Ezafe in question is inside the possessum, and the possessum (unlike the possessor) does not receive genitive case.

Meanwhile, D-forms of EZ are also used when a modified or a possessed noun [N-EZ MOD/POSS'R] is selected by adpositions. Some examples are given in (29). In (29a) the modified noun 'black table' and in (29b) the possessive construction 'your mother', appear as complements of an adposition; the Ezafe morpheme is then realized as *-dê/-da*. Once again, the *-dê/-da* morpheme reflects the phi-features of the head noun as well, with "table" being masculine and "mother" feminine.

- (29) a. [[*mase-dê* *siyay*] *sero*]
 table-DE black on
 'on the black table'
- b. [[*ma-r-da* *to*] *rê*]
 mother-OBL.F-DE 2SG.OBL to
 'to your mother'

¹⁸ We notice that the order in (28b) would require roll-up movement of the most embedded NP and its possessor together. This stands in contrast to (27b), which would involve spec-to-spec movement of the kind shown in figure 17. We speculate that this may be due to compound formation involved in the product name, rendering the head noun incapable of moving independently. Regardless, the point remains that it is the embedded genitive context that produces D-forms in Southern Zazaki.

It has been argued above that D-forms are the realization of Ezafe (on a modified or possessed noun) whenever it receives genitive case (GEN) in a possessive construction. We can extend this generalization to the cases involving adpositions by positing that adpositions assign genitive case to their complements in Zazaki. This is a common property cross-linguistically, including in related languages like Balochi (Jahani and Korn 2009), and other languages in the area, such as Turkish (Kornfilt 1997; Göksel and Kerslake 2005), among others. It should importantly be noted here that the appearance of the D-forms is dependent on GEN case and not simply any OBL case. This becomes clear when looking at modified nouns used in other OBL positions, as discussed in section 2, namely (7), (8), (9) and (11a). All of these examples involved [N-EZ MOD/POSS'R] used in an OBL position, and yet the regular form of Ezafe (rather than the d-form) appears. To summarize, we have proposed that the D-forms of Ezafe are used in Southern Zazaki whenever an Ezafe construction embedded in a genitive case context, either governed by the Poss head in a possessive construction or by an adposition.¹⁹

5.2 D-forms as “Double Ezafe”

Larson and Yamakido (2006) (also Larson and Samiian 2020) develop a theory of DP structure where all DP modifiers begin postnominally as the complements of D. In their account, all [+N] elements, including nominal modifiers such as adjectives, as well as nominal prepositions need to be case-licensed. They argue that Ezafe in Iranian languages is a case-assigning element which licences all [+N] elements in their base position. In their case-theoretic account of Ezafe, the D-forms of Ezafe in (Southern) Zazaki are taken to be instances of double case marking or case stacking. They suggest that when a phrase containing Ezafe is embedded in a larger Ezafe construction or when a phrase containing Ezafe is the complement of a postposition, the embedded Ezafe morpheme in (Southern) Zazaki appears as *-dê/-da* (depending on the phi-features of the head noun) as a reflex of the double Ezafe, i.e. double case marking (Plank’s 1995 suffixaufnahme). In other words, for Larson and Yamakido, in Zazaki, the D-forms are portmanteaus of Ezafe as a case marker and another second case marker, either an Ezafe or a general oblique case marker, i.e. they are instances of “double Ezafe”. Two examples from Larson and Yamakido are given in (30) and (31)²⁰ and their schematic analyses of these examples are shown in (32) and (33).

¹⁹ D-forms have also been observed in the context of allatives; e.g. the spatial goals of the verbs of directed motion such as *go*, *come* and *fall* are mostly encoded as DPs in the postverbal position. The Ezafe appears as *-dê/-da* when we have [NOUN-EZ- MOD/POSS'R] in this position ((i) and (ii)). We posit that these postverbal DPs are introduced by a silent adposition assigning GEN case.

(i) *Ez şî-ya baxçê-dê xo*
1SG.DIR go.PST-1SG garden-DE self
'I went to my garden'

(ii) *Ez şî-ya baxçê-dê xaseki*
1SG.DIR go.PST-1SG garden-DE beautiful
'I went to the beautiful garden'

²⁰ “Sez” stands for “second Ezafe”

- (30) [HEAD-EZ [HEAD-de/-da MOD]]
kutik-e [əmīryan-de ma]
 dog-EZ neighbor.OBL-SEZ us
 ‘our neighbor’s dog’

- (31) [HEAD-de/da MOD] P
[mar-da to] ra
 mom.OBL-SEZ you.OBL from
 ‘from your mother’

- (32) $[_{EXP} -e [_{DP} Pro D [_{D'} [_{NP} əmīryan] [_{D'} t [_{EXP} -de ma]]]]]$
 (33) $[_{PP} [_{DP} Pro D [_{D'} [_{NP} mar] [_{D'} t [_{EXP} -da to]]]] ra]$
- [OBL] [OBL]

Larson and Yamakido’s account faces a problem when we consider other cases of an Ezafe embedded under another Ezafe. We can see that the D-forms do not appear in all of these contexts, suggesting that these D-forms are more than “double” or “embedded” Ezafe. This is shown clearly in the contrast in (34)-(35). In (34), [HEAD-EZ MOD] is in a possessive relation with the head noun, hence receiving genitive case from the Poss head, whereas in (35), [HEAD-EZ MOD] is a complex modifier in an adjectival relation with the head noun (see also Todd 2002). The former context leads to the appearance of the d-form of Ezafe but the latter context does not.

- (34) [HEAD-EZ1 [HEAD-EZ2 MOD]] with [HEAD-EZ2 MOD] as possessor: EZ2 = DE
şit-ê manga-r-da siya
 milk-EZ.M.OBL cow-OBL.F-DE black
 ‘the black cow’s milk’
- (35) [HEAD-EZ1 [HEAD-EZ2 MOD]] with [HEAD-EZ2 MOD] as modifier: EZ2 ≠ DE
sol-ê siya-yê tari
 shoe-EZ.PL black-EZ.PL dark
 ‘the dark black shoes’

The above discussion undermines the “double Ezafe” analysis of *-dê/-da*, as suggested by Larson and Yamakido (2006) and Larson and Samiiian (2020) and crucially highlights the importance of the specific genitive case relation with the head noun. Only Ezafe constructions that receive GEN case will exhibit the appearance of the D-forms.²¹ Below, we develop an analysis that captures the distribution of the D-forms in Southern Zazaki.

²¹ There is an additional conceptual problem with Larson and Yamakido’s (2006) analysis of D-forms of Ezafe as case-stacking or suffixaufnahme, i.e. double Ezafe. In their general analysis, they treat Ezafe as a case assigner, whereas suffixaufnahme is a stacking of case markers, and hence not applicable to the Zazaki D-forms even if one follows their account of Ezafe generally. This is an un-addressed contradiction in Larson and Yamakido’s work.

5.3 D-forms as allomorphs of EZ

We have argued above that Southern Zazaki D-forms are used to realize Ezafe whenever the Ezafe construction receives genitive case, either by a Poss head in a possessive construction or by a (genitive case-assigning) adposition. The configurations for these two environments are given schematically in (36), where DE stands for both *-dê* and *-da*.

- (36) a. [_{NP} NOUN-EZ1 [_{NP-GEN} X-DE Y]
 b. [_{PP} [_{NP} NOUN-DE X] P_{GEN}]

Having established the contexts in which the D-forms of Ezafe appear in Southern Zazaki – namely an Ezafe construction embedded in a genitive-case context – we now seek to understand how best this variable morphology can be understood in linguistic terms. In particular, one can ask whether the use of D-forms of Ezafe should be treated similarly to the morphosyntactic allomorphy we discussed in sections 1 to 4. Recall that in those cases, we were dealing with different forms of Ezafe surfacing, depending on the phi- and case values of the Ezafe morpheme, obtained via nominal concord. Therefore, the *-o* form of Ezafe was analyzed as the realization of EZ with phi- valued as masculine singular and case valued as DIR. Similarly, the *-a* form of Ezafe was analyzed as the realization of EZ with phi- valued as feminine singular regardless of its case value. Finally, the *-ê* form of EZ is used in all other contexts, including all plural contexts but also masculine singular when its case value is OBL.

Can we conceptualize the appearance of the D-forms of Ezafe in Southern Zazaki along the same lines discussed above? The answer to this question is affirmative only with respect to the vowels used in the D-forms. The choice of *-ê/-a* forms solely depends on the phi- and case values of the EZ morpheme. For one, because the D-forms are always used in the genitive context, the case value is GEN(=OBL) and as such the vowel in these forms is realized as either *-a* (used with phi- valued as feminine singular) or *-ê* (used with phi- valued as plural or masculine singular). Importantly, the *-o* vowel is absent from the paradigm because its use is restricted to DIR case (with masculine singular features). Meanwhile, the consonantal part of the D-forms of EZ (namely [d]) cannot simply be analyzed as the realization of phi- and case values. Crucially, one does not come across the D-forms with simple cases of GEN, e.g. the Ezafe used in a simple possessive construction (see (10) above). Having established that the D-forms cannot be accounted for as straightforward cases of nominal concord, we can now turn to the proper analysis of these forms.

When we look at the D-forms (DE) of the EZ in Southern Zazaki, we see that they present a compelling case of grammatically conditioned allomorphy.²² Grammatically conditioned allomorphy refers to a type of contextual allomorphy where a morpheme takes on distinct forms in different syntactic contexts. Unlike phonologically conditioned allomorphy, which is triggered by phonological factors like surrounding sounds, syllable structure, and stress patterns, grammatically conditioned allomorphy is influenced by morphosyntactic features such as case, number, tense, or the syntactic position of a morpheme within the structure (Carstairs-McCarthy 2001; Embick and Noyer 2007; Embick 2010; Jonathan Bobaljik 2012; Alec Marantz 2013; Weisser 2019; Rolle 2023).

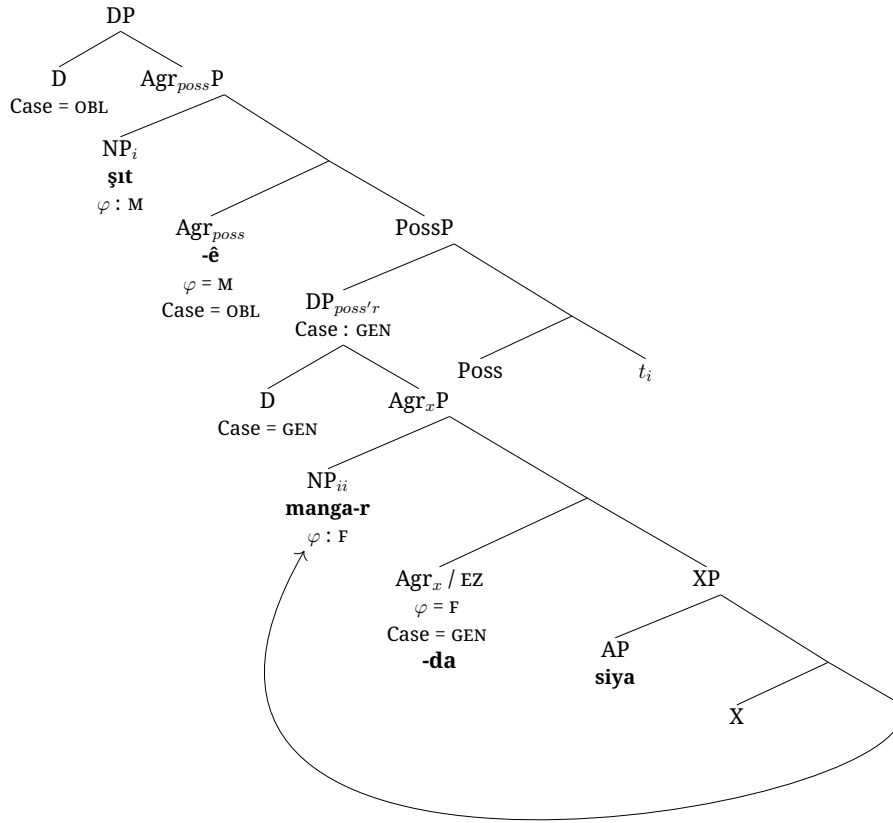
²² See Weisser (2019) for a systematic contrast between allomorphy and agreement.

In the case under discussion, the Ezafe forms in Southern Zazaki show outward sensitivity to the syntactic context they appear in. In grammatically conditioned allomorphy, like all cases of contextual allomorphy, locality plays a crucial role, i.e. a morpheme's form can only show sensitivity to syntactic contexts that are local to said morpheme. Meanwhile, the notion of locality has been defined distinctly by different authors. While locality is defined linearly by some (see, for example, Embick 2010), others have argued for a structural definition of locality (see, for example, Johnathan Bobaljik and Harley 2017, Weissner 2019). According to the latter view, the conditioning syntactic feature does not need to be linearly adjacent to the morpheme showing allomorphy. In the case of the Southern Zazaki Ezafe, as we will see more clearly in the structure and discussion below, the relevant structural relation is government, in line with Halle and Marantz's (1993) original proposal.

With this basic generalization about the appearance of *-dê/-da* in place, we can turn to the structure of examples we have considered so far. Let us start with an example of a modified noun used as the possessor in the possessive construction. Example (27a) is repeated in (37) with the corresponding tree structure given in (38) (some movement and agreement ignored for readability). As indicated in the structure presented in (38), the D-forms adhere to the structural locality constraints of grammatically conditioned allomorphy. In this configuration, as the D head c-commands Agr_xP , the Agr_x head is positioned within the government domain of the D head (à la Halle and Alex Marantz 1993), both being in the same maximal phrasal DP projection (Jonathan Bobaljik 2012). Recall that the EZ in our system is taken as the realization of the Agr heads with phi features valued with an Agree relation established with the NP and with the GEN case feature on D spreading onto the EZ marker inside the DP via case concord. The NP in the specifier position of the AgrP does not function as an intervener, as linear adjacency is irrelevant and specifiers are excluded from the structurally-defined locality domain (Johnathan Bobaljik and Harley 2017).

- (37) *şit-ê* *manga-r-da* *siya*
 milk-EZ.M.OBL cow-OBL.F-DE black
 'the black cow's milk'

(38)



Recall that the other two environments where D-forms appear are other instances where the Ezafe-containing DP is governed by a Genitive-assigning head, either when a possessive construction itself is a possessor, or when the Ezafe-containing DP is the complement of a Genitive-assigning adposition. Both of these constructions look very similar to the one presented in (38): the Ezafe (or Agr head) is either governed by the Poss head itself or an adposition which assigns genitive case, resulting in grammatically-conditioned allomorphy of the embedded Ezafe morpheme.²³ With space considerations in mind, we are leaving out the tree structures for these constructions.

We have so far seen the use of the D-forms of EZ in Southern Zazaki in contexts which involve GEN case, either assigned by a Poss head or an adposition. We conceptualized the distribution of the D-form as an allomorph of EZ, conditioned by the grammatical context of GEN case. Next, we consider another context in which D-forms of EZ are used in Southern Zazaki.

²³ Per Karim 2021, Zazaki D-forms have a distinct historical source from other EZ forms. While EZ is proposed to have descended from the Old Iranian relative pronoun *yat*, D-forms have their origins in the distal demonstrative pronoun *tat*. The distal demonstrative *tat* appeared when possessors became detached from the head noun syntactically. This dual diachronic source for the two forms of Ezafe provides indirect support for the allomorphy analysis of these elements.

5.4 D-forms in the context of indefinites

In this subsection, we consider the distribution of EZ in the context of indefinite noun phrases. While Southern Zazaki employs the D-forms of EZ in instances associated with GEN case, as shown above, we see that the use of D-forms in the context of indefinites is not confined solely to cases involving GEN case assignment.

First, it is worth pointing out that the indefinite marker in this dialect is *-ê*, as shown in (39).

- (39) a. *kutık-ê*
 ‘a dog’
 b. *kenek-ê*
 ‘a girl’

Some of the indefinite contexts can be associated with GEN case, which can be unified with the cases we considered in the previous sections. For example, when a modified indefinite noun phrase is selected by an adposition (40) or is used in a GEN context (41), the D-forms of EZ are used, as expected.²⁴

- (40) a. *mas-ên dê siya sero*
 table-INDF DE black on
 ‘on a black table’
 b. *mang-ên da siya ra*
 cow-INDF DE black from
 ‘from a black cow’

- (41) a. *mas-ên dê min*
 table-INDF DE 1SG.OBL
 ‘a table of mine’
 b. *mang-ên da min*
 cow-INDF DE 1SG.OBL
 ‘a cow of mine’

However, the use of the D-forms of EZ in the context of indefinites in Southern Zazaki is not limited to GEN-case contexts. As we can see in (42), D-forms of EZ are used in general indefinite contexts as well. Crucially, in this context, just as the D-forms discussed in previous sections, the choice of the vowel reflects the morphosyntactic features of the head noun, mirroring the paradigm we find with regular Ezafe. Thus, in (42a), we see the vowel *o* reflecting masculine singular phi-features and DIR case, and in 42b and 42d, we see *a*, which is the feminine singular form for both DIR and OBL cases. In 42c, the vowel is *ê*, which is the masculine singular OBL form.

²⁴ Barker (1998) analyzes English examples like (41) as double genitives: Barker suggests that double genitives are structurally equivalent to possessive partitives (like ‘two of John’s friends’), where a genitive-assigning head takes the embedded complex DP as a complement. If we apply this proposal to our Zazaki examples, once again, we have a functional head assigning Genitive case to an Ezafe-containing-DP, yielding *-dê/-da*.

- (42) a. *Kutık-ên do siya mın vin-en-o*
 dog-INDF DE black 1SG.OBL see-PRS-3SG.M
 ‘A black dog sees me.’
- b. *Mang-ên da siya mın vin-en-a*
 cow-INDF DE black 1SG.OBL see-PRS-3SG.F
 ‘A black cow sees me.’
- c. *Ez kutık-ên dê siya vin-en-a*
 1SG.DIR dog-INDF DE black see-PRS-1SG
 ‘I see a black dog.’
- d. *Ez mang-ên da siya vin-en-a*
 1SG.DIR cow-INDF DE black see-PRS-1SG
 ‘I see a black cow.’

It is worth noting the reappearance of the *-o* form in this context. Recall that the D-forms in the previous sections were never accompanied with *-o* as they were all GEN (or OBL) contexts, hence the DIR form was never employed. In the absence of a GEN context, the reappearance of the *-o* form, used only in DIR contexts, is not surprising. What is new here in contrast to definite contexts, is the additional *d-*.

We attribute the use of the D-forms of Ezafe in Southern Zazaki in the general context of indefinites to a more general ban on the co-occurrence of the regular form of Ezafe and the indefinite marker, also found in other Iranian languages²⁵. For instance, there is a blocking effect between the indefinite marker and EZ in Persian and Hawramî: in Persian, the Ezafe vowel blocks the indefinite *-i* (Ariyae 2023, p. 3), while in Hawrami the opposite occurs, yielding no overt EZ on indefinite nouns (S. A. Peters et al. 2023, p. 5).²⁶ Meanwhile, when we look at Southern Zazaki, we see that this variety develops a different strategy to address this incompatibility. Southern Zazaki solves the problem by avoiding the regular Ezafe and introducing the D-allomorphs of EZ (modern descendant of distal deictic *tat*), following the indefinite marker²⁷.

Let us look at the structure of a modified indefinite example repeated in (43), as shown in (44). In (44), we follow the analysis of Jambrovic and Kahnemuyipour (2023) of indefinites in Central Kurdish by positing the nominalizing layer (*nP*) as the locus for indefiniteness (following, in turn, Julien 2003; Julien 2005 on Scan-

²⁵ Two types of indefinite markers are used in Iranian languages, *yak* and *ew* (Karim 2021; Karim 2023). Central Kurdish, Northern Kurdish and Central Zazaki use *yak*, while New Persian, Hawrami, Southern Zazaki and Gilaki use *ew*. The incompatibility seems to apply only to the indefinite markers of the *ew* type.

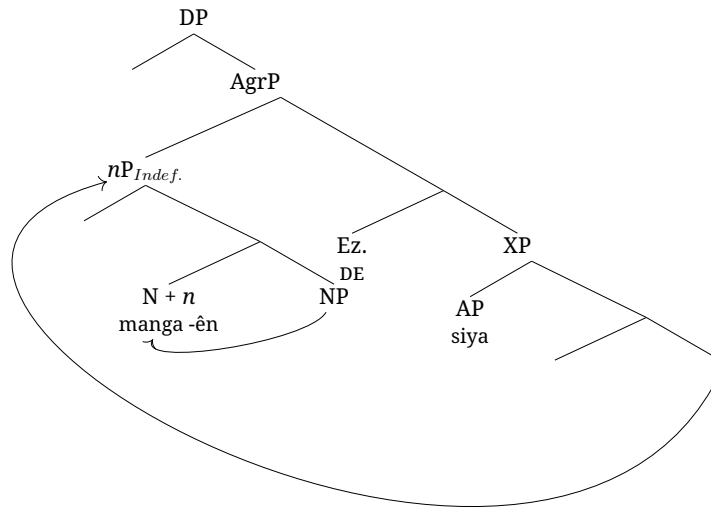
²⁶ Recall that the regular form of Ezafe is a descendant of the Old Iranian relative pronoun *yat*, which may have acted as a definite article. Therefore, the ban on the co-occurrence of the indefinite marker and Ezafe may be diachronically rooted in the incompatibility of definite and indefinite forms.

²⁷ An anonymous reviewer suggests that we can analyze the D-forms of Ezafe in Southern Zazaki as allomorphs of Ezafe which appear in contexts where Ezafe cannot attach to the head noun directly. Put differently, according to this proposal, we find the D-forms whenever there is some morphology intervening the head noun and the Ezafe. This proposal is very much in the spirit of what we are suggesting for the D-forms used in the context of indefinites. However, we have not extended this analysis to the other cases of D-forms in the previous sections, as in those contexts such morphology seemed to be absent at least in several cases, see for example (25b), (26a), (29a), and the allative examples in footnote 19. Moreover, sometimes the regular form of Ezafe is used even when there is intervening morphology, as in (28b). We conclude that resorting to the genitive context is needed for the correct analysis of the D-forms discussed in the previous sections.

dinavian). Thus, the head noun moves to the head of *nP*, which itself moves into spec,AgrP, just as in the analyses discussed above.

- (43) *mang-ên da siya*
 cow-INDEF DE black
 ‘a black cow’

(44)



To summarize, we propose in this section that in Southern Zazaki the D-allomorph of EZ is employed in two disjoint environments:²⁸ the GEN case and indefinites.²⁹

²⁸ The use of allomorphs in disjoint grammatical contexts is not unusual. For example, in English, allomorphs such as [ʃelv] or [θiv] are used both in the context of plural (*shelves, thieves*), as well as to form verbs (*shelve, thief*).

²⁹ In another Zazaki dialect, namely Northern Zazaki, when an indefinite noun is modified by an adjective, we see the expected forms of EZ (showing sensitivity to phi and case) followed by an invariant form –dı (ia) or –de in some varieties. In (ia), we see an example of a masculine noun modified by an adjective used in a DIR case position, hence the EZ form is –o. With the feminine noun in (ib), the EZ form is –a and when the masculine nouns is used in an OBL position, we get –ê (ic). This pattern reported by Werner (2018) was confirmed by a native speaker consultant, but some authors (namely Keskin (2010, p. 30) and Paul (1998, p. 186)) report a pattern similar to Southern Zazaki for Northern Zazaki.

(i) Northern Zazaki indefinites

- a. *kutık-o dı gırs*
 dog-EZ.M.DIR DE big
 ‘a big dog’
- b. *kenek-a dı rındeķı*
 girl-EZ.F DE beautiful
 ‘a beautiful girl’
- c. *kutık-ê dı mun*
 dog-EZ.M.OBL DE 1SG.OBL
 ‘a dog of mine’

6 Conclusions

The foregoing discussion has aimed at describing two puzzles in the expression of the Ezafe in Southern Zazaki. This language, like many other Northwestern Iranian languages, has a relatively rich paradigm expressing both phi-features and case on the linking Ezafe morpheme in the nominal domain. However, certain unexpected deviations warranted further investigation. In the process of accounting for these puzzles, we have described the morphosyntax of an agreeing Ezafe with machinery that has previously been successful in accounting for invariant linking forms in other Iranian languages. Namely, we have applied the agree-based movement approaches of Kahnemuyipour (2014) and Atlamaz (2016) to the Zazaki case, and have shown how they can account for the Southern Zazaki pattern with some well-grounded assumptions.

The first puzzle we approached was that of case expression on the Zazaki Ezafe. This linker morpheme, under normal circumstances, reflects the case value that the entire DP receives. For instance, when the DP resides in a DIR case environment, DIR forms of an agreeing Ezafe appear, which can be explained with machinery that has previously been successful in accounting for invariant linking forms in other Iranian languages coupled with nominal concord. Meanwhile, it was shown that when a possessor appears in the DP, OBL forms of the Ezafe are used invariably, i.e. even in a DIR environment. We argued that what differentiates the possessive environment from the previous is merely the fact that in this context, Case is valued internally. Under normal circumstances, Case Concord extends whatever external case the DP receives throughout the nominal structure. Through the same mechanism, when an element enters the DP already bearing case (as it is with possessors, receiving genitive), the case on all heads (including D) becomes valued as GEN/OBL.

The second puzzle involved the appearance of the D-forms of Ezafe in certain contexts in Southern Zazaki. Specifically, we showed that the Ezafe, when employed in a phrase that is itself governed by the genitive case, takes on a separate paradigm – the D-forms (*-dê/-da*). Previous analyses simply described the phenomenon, or analyzed it as case stacking / suffixaufnahme. We noted, importantly, that the D-forms do not appear just whenever Ezafe is embedded (and would have a double or “stacked” case), but that, for one, D-forms appear under certain postpositions, and secondly do not appear when the Ezafe-bearing nominal is used as a modifier of another phrase, and not the possessor. We analyzed the D-forms as grammatically-conditioned allomorphs of Ezafe appearing in the context of genitive, with the assumption that the postpositions also assign genitive case to their complements. This allomorphy analysis required no extra theoretical machinery, and no modification of the prior analysis of the morphosyntax of Ezafe constructions to account for the distribution of D-forms in Southern Zazaki. We further argued that the use of the D-forms of Ezafe in the context of indefinites is another instantiation of contextual

What can be said about the pattern illustrated in (i)? It is important to note that the D-forms of EZ are not otherwise used in Northern Zazaki. It seems that Northern Zazaki employs another strategy to address the incompatibility of indefinite and EZ. When contending for the same morphological slot after the head noun, EZ occupies this slot, eliminating the indefinite marker and introducing the resumptive D-form representing the indefinite marker (see also Werner (2018)). This raises a question about the structure of the indefinite noun phrase in Northern Zazaki. We tentatively take the indefinite structure in both Northern Zazaki and Southern Zazaki to be identical, and the reordering of the EZ and indefinite markers to be the result of a local dislocation operation (Embick 2007).

allomorphy which is the result of the incompatibility between the regular form of Ezafe and the indefinite marker, also found in other Iranian languages.

While there are certain lingering questions with respect to the Southern Zazaki Ezafe, in particular as it pertains to its use in the context of indefinites, this study has provided the first fulsome explanation of the morphosyntax of Ezafe in this language, taking into account deviations from the expected Ezafe paradigm that prove enlightening for a theory of the nominal domain in Iranian languages and beyond.

References

- Abels, Klaus and Ad Neeleman (2012). “Linear Asymmetries and the LCA.” In: *Syntax* 15, pp. 25–74.
- Ariyae, Koorosh (2023). “Nominal linkers in Persian.” In: *Toronto Working Papers in Linguistics* 45.1. doi: <https://doi.org/10.33137/twpl.v45i1.41683>.
- Atlamaz, Ümit (2016). “Merging modifiers of an NP before its arguments.” In: *Proceedings of the 33rd West Coast Conference on Formal Linguistics*. Somerville: Cascadilla Proceedings project, pp. 47–56.
- Babby, Leonard (1987). “Case, prequantifiers, and discontinuous agreement in Russian.” In: *Natural Language & Linguistic Theory* 5.1, pp. 91–138.
- Baier, Nico (2015). “Adjective agreement in Noon: Evidence for a split theory of noun-modifier concord.” In: *Proceedings of the 45th Annual Meeting of the North East Linguistic Society*. Ed. by Thuy Bui and Deniz Özyıldız. Amherst, MA: GLSA Publications, pp. 67–80.
- Baker, Mark (2008a). *The Syntax of Agreement and Concord*. Cambridge: Cambridge University Press.
- Baker, Mark (2008b). *The Syntax of Agreement and Concord*. Cambridge: Cambridge University Press.
- Barker, Chris (1998). “Partitives, double genitives and anti-uniqueness.” In: *Natural Language and Linguistic Theory* 16, pp. 679–717.
- Bejar, Susana and Diane Massam (1999). “Multiple Case Checking.” In: *Syntax* 2.2, pp. 65–79.
- Bjorkman, Bronwyn and Hedde Zeijlstra (2019). “Checking up on (ϕ -)Agree.” In: *Linguistic Inquiry* 50.3, pp. 527–569.
- Bobaljik, Johnathan and Heidi Harley (2017). “Suppletion is local: Evidence from Hiaki.” In: *The structure of words at the interfaces*. Vol. 68, pp. 141–162.
- Bobaljik, Jonathan (2012). *Universals in comparative morphology: Suppletion, superlatives, and the structure of words*. MIT Press.
- Bonet, Eulàlia (2013). “Agreement in two steps (at least).” In: *Distributed morphology today: Morphemes for Morris Halle*. Ed. by O. Matushansky and A. Marantz. Cambridge, MA: MIT Press.
- Brattico, Pauli (2008). “Kayne’s model of Case and Finnish nominal phrases.” In: *Nordic Journal of Linguistics* 31, pp. 135–160.
- Brattico, Pauli (2010). “The two-part models and one-part models of nominal case: evidence from case distribution.” In: *Journal of Linguistics* 45, pp. 47–81.
- Brattico, Pauli (2011). “Case assignment, case concord, and the quantificational case construction.” In: *Lingua* 121, pp. 1042–1066.
- Carstairs-McCarthy, A. (2001). “Grammatically conditioned allomorphy, paradigmatic structure, and the Ancestry Constraint.” In: *Transactions of the Philological Society* 99, pp. 223–245.
- Carstens, Vicki (2000). “Concord in minimalist theory.” In: *Linguistic Inquiry* 31.2, pp. 319–355.
- Carstens, Vicki (2001). “Multiple agreement and case deletion: Against ϕ -incompleteness.” In: *Syntax* 4.3, pp. 147–163.
- Carstens, Vicki (2011). “Hyperactivity and hyperagreement in Bantu.” In: *Lingua* 121.5, pp. 721–741.
- Carstens, Vicki (2016). “Delayed Valuation: a reanalysis of “upwards” complementizer agreement and the mechanics of Case.” In: *Syntax* 19.1, pp. 1–42.

- Chomsky, Noam (1981). *Lectures on Government and Binding*. Dordrecht, Foris Publications.
- Chomsky, Noam (1986). *Barriers*. MIT Press.
- Chomsky, Noam (2000). “Minimalist Inquiries: The Framework.” In: *Step by Step*. Ed. by Roger Martin, David Michaels, and Juan Uriagereka. Cambridge, Mass.: MIT Press, pp. 89–155.
- Chomsky, Noam (2001). “Derivation by Phase.” In: *Ken Hale: A life in Language*. Ed. by M. Kenstowicz. Cambridge: MIT Press, pp. 1–52.
- Chomsky, Noam (2013). “Problems of Projection.” In: *Lingua* 130, pp. 33–49.
- Cinque, Guglielmo (1994). “On the Evidence for Partial N Movement in the Romance DP.” In: *Paths towards Universal Grammar: Studies in Honor of Richard S. Kayne*. Ed. by G. Cinque, J. Koster, J.-Y. Pollock, L. Rizzi, and R. Zanuttini, pp. 85–110.
- Cinque, Guglielmo (2002). *The Functional Structure of DP and IP – The Cartography of Syntactic Structures*. Vol. 1. Oxford: Oxford University Press.
- Cinque, Guglielmo (2005). “Deriving Greenberg’s Universal 20 and its Exceptions.” In: *Linguistic Inquiry* 36, pp. 315–332.
- Cinque, Guglielmo (2010). *The Syntax of Adjectives*. MIT Press, Cambridge, MA.
- Collins, Christopher (2004). “The agreement parameter.” In: *Triggers*. Ed. by A. Breitharth and H. van Riemsdijk. Mouton de Gruyter, pp. 115–136.
- Danon, Gabi (2011). “Agreement and DP-internal feature distribution.” In: *Syntax* 14.4, pp. 297–317.
- Delsing, Lars-Olaf (1993). “The internal structure of noun phrases in the Scandinavian languages.” PhD thesis.
- den Dikken, Marcel (2006). *Relators and Linkers: The Syntax of Predication, Predicate Inversion and Copulas*. MIT Press, Cambridge, Massachusetts.
- Embick, David (2007). “Linearization and local dislocation: Derivational mechanics and interactions.” In: *Linguistic Analysis* 33.3-4, pp. 303–336.
- Embick, David (2010). *Localism versus globalism in morphology and phonology*. Cambridge, MA: MIT Press.
- Embick, David and Rolf Noyer (2007). “Distributed Morphology and the Syntax-Morphology Interface.” In: *The Oxford Handbook of Linguistic Interfaces*. Ed. by Gillian Ramchand and Charles Reiss. Oxford, UK: Oxford University Press, pp. 289–324.
- Ghaniabadi, Saeed (2010). “The Empty Noun Construction in Persian.” PhD Dissertation, University of Manitoba, Winnipeg MB.
- Ghomeshi, Jila (1997). “Non-projecting Nouns and the Ezafe Construction in Persian.” In: *Natural Language and Linguistic Theory* 15, pp. 729–788.
- Giusti, Giuliana (2008). “Agreement and concord in nominal expressions.” In: *The Bantu-Romance Connection*. Ed. by C. De Cat and K. Demuth. John Benjamins, pp. 201–237.
- Göksel, Asli and Celia Kerslake (2005). *Turkish: a Comprehensive Grammar*. London; New York: Routledge.
- Gündoğdu, Songül and Amine Bulan (2023). “Ezafe in Zazaki.” In: *Toronto Working Papers in Linguistics* 45.1. DOI: <https://doi.org/10.33137/twpl.v45i1.41679>.
- Halle, Morris and Alex Marantz (1993). “Distributed Morphology and the Pieces of Inflection.” In: *The View from Building 20*. Ed. by Ken Hale and Samuel Jay Keyser. Cambridge, MA: MIT Press, pp. 111–176.
- Halpert, Claire (2015). *Argument Licensing and Agreement*. Oxford University Press.

- Harley, Heidi (2013). "Feature matching and case/number dissociation in Hiaki." In: *Revista Linguistica* 9, pp. 1–9.
- Jahani, Carina and Agnes Korn (2009). "Balochi." In: *The Iranian Languages*. Ed. by Gernot Windfuhr. New York: Routledge, pp. 634–692.
- Jambrovic, Samuel and Arsalan Kahnemuyipour (2023). "Number, Definiteness and Compounding in Central Kurdish." In: *North American Conference in Iranian Linguistics*.
- Julien, Marit (2003). "Double Definiteness in Scandinavian." In: *Nordlyd* 31.1, pp. 230–244. DOI: <https://doi.org/10.7557/12.27>.
- Julien, Marit (2005). *Nominal phrases from a Scandinavian perspective*. Amsterdam: John Benjamins. DOI: <https://doi.org/10.1075/la.87>.
- Kahnemuyipour, Arsalan (2014). "Revisiting the Persian Ezafe Construction: A Roll-up Movement Analysis." In: *Lingua* 150, pp. 1–24.
- Kahnemuyipour, Arsalan and Andrew Peters (2019). "Separating Concord and Agree: The Case of Zazaki Ezafe." In: *Proceedings of the CLA Annual Meeting*.
- Karim, Shuan (2021). "The synchrony and diachrony of New Iranian nominal morphosyntax." PhD thesis. The Ohio State University.
- Karim, Shuan (2023). "Ezafe and the article." In: *Advances in Iranian Linguistics II*. Ed. by Simin Karimi, Narges Nematollahi, Roya Kabiri, and Jian Gang Ngui, pp. 130–153.
- Karimi, Simin (2008). "Raising and Control in Persian." In: *Aspects of Iranian Linguistics*. Ed. by S. Karimi, V. Samiiian, and D. Stilo. Cambridge Scholars Publishing Co.
- Karimi, Yadgar (2007). "Kurdish Ezafe construction: Implications for DP structure." In: *Lingua* 117, pp. 2159–2177.
- Keskin, Mesut (2010). "Zazaca üzerine notlar (Notes on Zazaki)." In: *Herkesin Bildiği Şır: Dersim*. Ed. by Şükrü Aslan. İletişim, pp. 221–244.
- Kornfilt, Jaklin (1997). *Turkish*. London: Routledge.
- Kramer, Ruther (2009). "Definite markers, phi-features, and agreement: amorphosyntactic investigation of the Amharic DP." PhD thesis. University of California, Santa Cruz.
- Landau, Idan (2016). "DP-internal semantic agreement: A configurational analysis." In: *Natural Language and Linguistic Theory* 34.3, pp. 975–1020.
- Larson, Richard and Vida Samiiian (2020). "The Ezafe Construction Revisited." In: *Advances in Iranian Linguistics*. Ed. by Richard Larson, Sedigheh Moradi, and Vida Samiiian. John Benjamins, pp. 173–236.
- Larson, Richard and Hiroko Yamakido (2006). "Zazaki "Double Ezafe" as Double Case-marking." In: *Paper presented at the LSA Meeting*.
- Larson, Richard and Hiroko Yamakido (2008). "Ezafe and the Deep Position of Nominal Modifiers." In: *Adjectives and Adverbs: Syntax Semantics and Discourse*. Oxford University Press, pp. 43–70.
- Marantz, Alec (2013). "Locality domains for contextual allomorphy across the interfaces." In: *Distributed morphology today: Morphemes for Morris Halle*, pp. 95–115.
- Matushansky, Ora (2008). "A case study of predication." In: *Studies in Formal Slavic Linguistics: Contributions from Formal Description of Slavic Languages* 6.5. Ed. by F. Marušič and R. Žaucer. Peter Lang, pp. 213–239.
- Norris, Mark (2014). "A theory of nominal concord." PhD thesis. University of California, Santa Cruz.

- Norris, Mark (2017a). "Description and analyses of nominal concord (Pt I)." In: *Language and Linguistics Compass* 11 (11).
- Norris, Mark (2017b). "Description and analyses of nominal concord (Pt II)." In: *Language and Linguistics Compass* 11 (11). URL: <https://doi.org/10.1111/lnc3.12267>.
- Norris, Mark (2018). "Unmarked case in Estonian nominals." In: *Natural Language & Linguistic Theory* 36, pp. 523–562.
- Paul, Ludwig (1998). *Zazaki: Grammatik und Versuch einer Dialektologie*. Wiesbaden: Dr. Ludwig Reichert Verlag.
- Paul, Ludwig (2009). "Zazaki." In: *The Iranian Languages*. Ed. by Gernot Windfuhr. Routledge, pp. 545–586.
- Pesetsky, David and Esther Torrego (2007). "The syntax of valuation and the interpretability of Features." In: *Phrasal and clausal architecture: Syntactic derivation and interpretation. In honor of Joseph E. Emonds*. Ed. by Simin Karimi, Vida Samiian, and Wend K. Wilkins. Amsterdam: John Benjamins, pp. 262–294.
- Peters, Sable Andrew, Zaniar Naghshbandi, and Arwin Rasouli (2023). "The Ezafe in Hawrami." In: *Toronto Working Papers in Linguistics* 45.1. DOI: <https://doi.org/10.33137/twpl.v45i1.41680>.
- Plank, Frans (1995). "(Re-)introducing Suffixaufnahme." In: *Double Case. Agreement by Suffixaufnahme*. Ed. by Frans Plank. Oxford University Press, pp. 3–110.
- Polinsky, Maria (2016). "Agreement in Archi from a minimalist perspective." In: *Archi: Complexities of Agreement in a Cross-Theoretical Perspective*. Ed. by M. Chumakina O. Bond G. Corbett and D. Brown.
- Richards, Norvin (2013). "Lardil "Case Stacking" and the Timing of Case Assignment." In: *Syntax* 61.1, pp. 42–76.
- Rolle, N. (2023). "Inward and Outward Allomorph Selection." In: *The Wiley Blackwell Companion to Morphology*. Ed. by Peter Ackema, Sabrina Bendjaballah, Eulàlia Bonet, and Antonio Fábregas.
- Samiian, Vida (1983). "Origins of Phrasal Categories in Persian." PhD Dissertation, UCLA.
- Samiian, Vida (1994). "The Ezafe Construction: Some Implications for the Theory of X-bar Morphosyntax." In: *Persian Studies in North America*. Ed. by M. Marashi. Bethesda, MD: Iranbooks.
- Samvelian, Pollet (2007). "A (Phrasal) Affix analysis of the Persian Ezafe." In: *Journal of Linguistics* 43.3, pp. 605–645.
- Samvelian, Pollet (2008). "The ezafe as a head-marking inflectional suffix: Evidence from Persian and Kurmanji Kurdish." In: *Aspects of Iranian Linguistics*. Ed. by Simin Karimi, Vida Samiian, and Donald Stilo. Cambridge Scholars Press, pp. 339–362.
- Shlonsky, Ur (2004). "The form of Semitic Noun Phrases." In: *Lingua* 114.12, pp. 1465–1526.
- Shlonsky, Ur (2010). "The Cartographic Enterprise in Syntax." In: *Language and Linguistics Compass* 4.6, pp. 417–429.
- Todd, Terry Lynn (2002). *A Grammar of Dimili. Also Known as Zaza*. Iremet Forlag, Stockholm.
- Toosarvandani, Maziar and Coppe van Urk (2014). "The syntax of nominal concord: What Ezafe in Zazaki shows us." In: *Proceedings of NELS*. Vol. 43.2, pp. 209–220.
- Weisser, Philipp (2019). "Telling Allomorphy from Agreement." In: *Glossa: a Journal of General Linguistics* 4.1. DOI: <https://doi.org/10.5334/gjgl.803>.

- Werner, Brigitte (2018). "Forms and Meanings of the Ezafe in Zazaki." In: *Endangered Iranian Languages*. Ed. by Saloumeh Gholami. Reichert Verlag.
- Zeijlstra, Hedde (2012). "There is only one way to agree." In: *Linguistic Review* 29.2, pp. 491–539.